

**Villanova University**

**Department of Mechanical Engineering**

**ME 4850 SPRING 2026**

**FINAL REPORT 3. THERMAL FLUID DESIGN OF A DATA CENTER  
RACK LIQUID COOLING SYSTEM**

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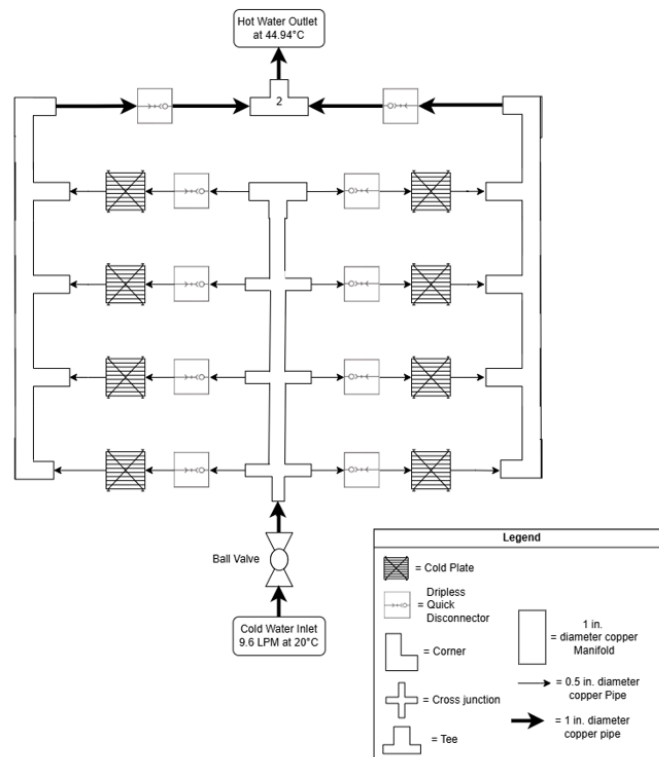
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## 1. Introduction and Specifications

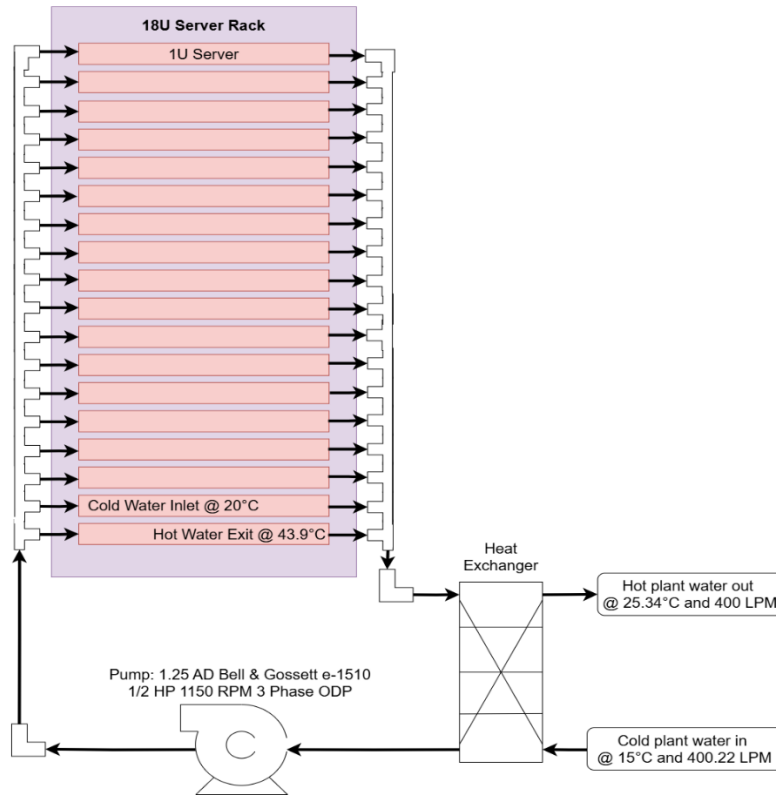
The previous report focused on the design of a server rack flow loop to deliver cooling water to cold plates that were cooling computer chips. That design resulted in a significant change in temperature and pressure loss throughout the system. This final report aimed to design the Coolant Distribution Unit (CDU) that cooled the water returning from the server rack and supplied the necessary head to overcome the system pressure losses. The heat exchanger design was for a water-to-water exchanger, cooling the hot water from the rack with cold water supplied by the plant.

## 2. Coolant Distribution Units Requirements

The Final Report was focused on designing a brazed-plate heat exchanger, a type of counter-flow heat exchanger that funnels flow into narrow channels. The channels alternated between the system's hot wastewater and plant-supplied cooling water. The next step was to select a pump that could operate in the system at peak efficiency, low-power, and the desired flow rate.



**Figure 1. P&ID of Individual Server Flow Loop**



**Figure 2. P&ID of the Entire System**

| <b>Fittings</b>                     | <b>Material</b> | <b>Diameter (in)</b> | <b>Length (in)</b> | <b>Quantity (#)</b> | <b>Price per unit (\$)</b> | <b>Part Number</b>                                        | <b>Site</b>          |
|-------------------------------------|-----------------|----------------------|--------------------|---------------------|----------------------------|-----------------------------------------------------------|----------------------|
| Elbow (90°)                         | Copper          | 2                    | 5                  | 2                   | 82.73                      | PFXPP9K                                                   | PROFLO               |
| Tee                                 | Copper          | 1.5                  | 5                  | 18                  | 76.99                      | PFXPPTJ                                                   | PROFLO               |
| Ball Valve                          | Stainless Steel | 1.5                  | 7.59               | 18                  | 480.00                     | SS-67TF24                                                 | Swagelok             |
| Dripless Quick Disconnect Set       | Stainless Steel | 0.5                  | 2                  | 180                 | 1812.00                    | SS-QDF8 / SS-QDC8                                         | Swagelok             |
| <b>Manifolds</b>                    |                 |                      |                    |                     |                            |                                                           |                      |
| Center Server Flow Loop Manifold    | Copper          | 1.5                  | 14                 | 1                   | Quote required             | MANI_1                                                    | Designed             |
| Perimeter Server Flow Loop Manifold | Copper          | 1.5                  | 16                 | 2                   | Quote required             | MANI_2                                                    | Designed             |
| Inlet/Outlet Server Rack Manifold   | Copper          | 2.5                  | 31.5               | 2                   | Quote required             | MANI_3                                                    | Designed             |
| <b>Components</b>                   |                 |                      |                    |                     |                            |                                                           |                      |
| Cold Plate                          | Copper          | N/A                  | 2                  | 144                 | Quote required             | COLDPLATE_1                                               | Designed             |
| <b>Piping</b>                       |                 |                      |                    |                     |                            |                                                           |                      |
| Copper Tube                         | Copper          | 0.5                  | 120                | 5                   | 43.17                      | 5175K123                                                  | McMaster             |
| Copper Tube                         | Copper          | 1                    | 120                | 3                   | 102.92                     | 5175K126                                                  | McMaster             |
| Hosing                              | Flexible PVC    | 2                    | 24                 | 1                   | 35.79                      | ZUSA-HT-3037                                              | Grainger             |
| <b>CDU</b>                          |                 |                      |                    |                     |                            |                                                           |                      |
| Heat Exchanger                      | Stainless Steel | N/A                  | 9.37               | 1                   | Quote required             | HTX_1                                                     | Designed             |
| Pump                                | Cast Iron       | 1.25 (at outlet)     | 32                 | 1                   | 4,318.95                   | 1.25 AD Bell & Gossett e-1510 1/2 HP 1150 RPM 3 Phase ODP | National Pump Supply |

**Figure 3. Full System Bill of Materials**

The Interim Report calculated the necessary inlet and outlet temperatures of the coolant as well as the desired operating flow rate. These design parameters are shown below in Table 1. Also shown in the table are the final results for the plant-supplied water and the total system impedance.

**Table 1. CDU Specifications**

| Parameter                                                 | Value     |
|-----------------------------------------------------------|-----------|
| Flow Rate of Hot Water                                    | 172.8 LPM |
| Flow Rate of Cold Water                                   | 400 LPM   |
| Water temperature entering HTX                            | 43.93°C   |
| Water temperature out of HTX and into the pump            | 20°C      |
| Required plant chilled water temperature entering the HTX | 15°C      |
| Temperature of the plant chilled water exiting the HTX    | 25.34°C   |
| Total System Impedance                                    | 6.6 psi   |

### 3. Water-to-Water Heat Exchanger (HTX) Design

One objective of this report was to design a brazed plate heat exchanger to cool the liquid water that was used in the server rack to cool the computer chips. The water was cooled by cold water supplied by the plant. Several important assumptions were made in this design. First, the plates in the heat exchanger were assumed to be smooth plates. Second, they were assumed to be experiencing constant heat flux. Third, the system was designed to operate at steady state.

The first step was to determine how much heat needed to be transferred by the system ( $q$ ). This was calculated using an energy balance equation, since the inlet temperature ( $T_i$ ) and outlet temperature ( $T_o$ ) of the water were design specifications from previous reports. The heat transfer equation is shown below, where  $c_p$  was the specific heat of water and  $\dot{m}$  was the mass flow rate of water, converted from the operating flow rate.

$$q = \dot{m} * c_p * (T_o - T_i) \quad (1)$$

The calculated total heat transferred was 288kW, at a mass flow rate of 2.88 kg/s and a total temperature change of 23.93°C. This total energy determined the necessary inlet water mass flow

rate. The main temperature consideration for the plant water was the water temperature that returned to the plant's cooling tower. Industry standards for this claimed that water should not return above 25°C. The inlet temperature was selected to be 15°C, which was shown to be a reasonable temperature for water supplied from cooling towers. Then, Equation 1 was rearranged to solve the mass flow rate of the cooling water from the plant (6.67 kg/s).

The next stage of design was to determine the heat exchanger effectiveness, which would in turn determine the Number of Transfer Units (NTU) needed to cool the water. The effectiveness was a ratio of the actual heat transfer rate ( $q$ ) and the theoretical maximum heat transfer rate ( $q_{max}$ ).

$$\varepsilon = q/q_{max} \quad (2)$$

The maximum heat transfer rate occurred when the hot water was cooled from its inlet temperature ( $T_{h,i}$ ) to the inlet temperature of the cooling water, which would be the maximum temperature change it could experience. This maximum change was represented in the following equation, where  $C_{min}$  is the minimum thermal capacity rate between the two exchanged liquids based on their mass flow rates and specific heats. Since both liquids were water, the specific heats were the same.  $C_{min}$  was dependent on the liquid with the lower flow rate. The calculation for the thermal capacity rate is also shown below.

$$q_{max} = C_{min}(T_{h,i} - T_{c,i}) \quad (3)$$

$$C_{min} = \dot{m} * c_p \quad (4)$$

The next calculation required was the thermal capacity rate ratio ( $C_r$ ), which was used with the effectiveness to determine the necessary NTU from a crossflow heat exchanger chart [A-1]. Based on the  $C_r$  and the effectiveness ( $\varepsilon$ ), the NTU was determined to be 1.5.

$$C_r = \frac{C_{min}}{C_{max}} \quad (5)$$

The next design goal was to determine the average convection coefficient ( $\bar{h}$ ). For this calculation, the channels of the heat exchanger were assumed to be straight, smooth channels in laminar flow with an infinite aspect ratio (base to height of the cross-section). The laminar flow assumption was then validated by calculating the Reynolds number through the channel. The result was a Nusselt correlation of a constant value, 8.24. Equation 6 calculated the Reynolds' number ( $Re$ ) as a function of the water velocity ( $V$ ), the kinematic viscosity ( $\nu$ ), and  $D_h$ .

$$Re = \frac{V * D_h}{\nu} \quad (6)$$

Equation 7 calculated the hydraulic diameter ( $D_h$ ) as a function of the heat exchanger's channel base ( $B$ ) and width ( $w$ ).

$$D_h = \frac{4Bw}{2B + 2w} \quad (7)$$

The calculated Reynolds number was 819 for hot water at the operating flow rate of 172.8 LPM, below the turbulent transition period at around 2000. Therefore, the assumed Nusselt value held for the operating point. Equation 8 calculated the water's convection coefficient ( $h$ ) as a function of the water's thermal conductivity ( $k$ ), the Nusselt correlation ( $Nu$ ), and  $D_h$ .

$$\bar{h} = \frac{Nu * k}{D_h} \quad (8)$$

Equation 9 calculated the necessary heat exchanger surface area ( $A$ ) to achieve the proper heat exchange rate based on the calculated number of transfer units ( $NTU$ ), the water's average convection coefficient ( $\bar{h}$ ), and  $C_{min}$ . The  $NTU$  was determined using the effectiveness- $NTU$  chart for counter-flow heat exchangers found in Appendix A-1 and the previously calculated  $C_r$  and  $\varepsilon$  values.

$$A = \frac{NTU * C_{min}}{\bar{h}} \quad (9)$$

The next step was to calculate the heat exchanger impedance to find the pressure drop contribution to the total system and inform pump selection. Equation 10 calculated the fanning friction factor ( $f_f$ ) as a function of  $Re$  and an exponential expansion of the heat exchanger's base-to-width ratio ( $\frac{b}{w}$ ) with experimentally determined coefficients.

$$f_f = \frac{1}{Re} [24 - 32.527 \left(\frac{b}{w}\right) + 46.721 \left(\frac{b}{w}\right)^2 - 40.829 \left(\frac{b}{w}\right)^3 + 22.954 \left(\frac{b}{w}\right)^4 - 6.089 \left(\frac{b}{w}\right)^5] \quad (10)$$

The pressure drop through the heat exchanger was due to both major and minor head losses. The total flow rate affected the friction in the pipes, resulting in major head loss ( $P_M$ ) and a pressure drop through the chosen fittings, resulting in minor head loss ( $P_m$ ). The major head loss, or pipe friction, was calculated by Equation 11 as a function of the friction factor ( $f$ ), the water velocity ( $V$ ), the water's density ( $\rho$ ), and a ratio of the pipe length ( $L$ ) to hydraulic diameter ( $D_h$ ).

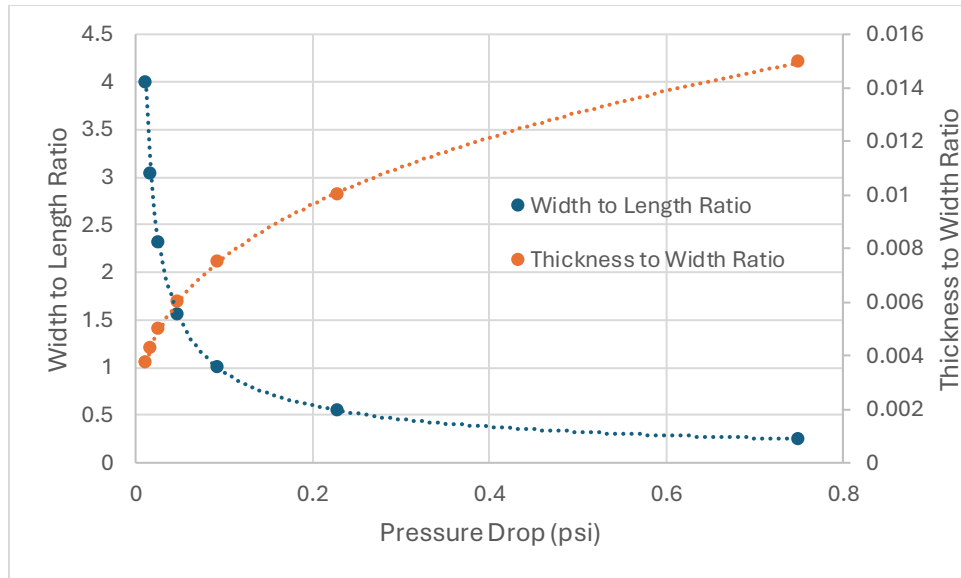
$$\Delta P_M = 4f_f\left(\frac{L}{D_h}\right)\left(\frac{\rho V}{2}\right) \quad (11)$$

The calculation for the minor loss was a function of the loss coefficient ( $K$ ), a factor dependent on the type of flow restriction. The friction factors for the straight pipe were found using the Moody Chart [A-2], based on the calculated Reynolds number in each pipe.

There was a head loss due to the distribution of the water from the through-pipes into the plate channels. This loss was assumed to be similar to a tee-branch, and a  $K$  value of 0.6 was used.<sup>1</sup> These loss coefficients were used in the minor head-loss ( $P_m$ ) Equation 12 calculation, which also accounted for the fluid's density and velocity.

$$\Delta P_m = K\left(\frac{\rho V}{2}\right) \quad (12)$$

With the heat exchanger's necessary area and related pressure drop calculated, the dimensions were determined. The number of plates, length, width, base length, thickness, and distance between the plates were determined to meet the required surface area. The main consideration was ensuring that the surface area met the requirement; the second consideration was evaluating how other factors could influence certain proportions. One factor was the pressure drop. The equations above demonstrated that the pressure drop was influenced by the hydraulic diameter and the channel length, and changing the exact distribution of the length (whether that be making the channels taller and thinner, or shorter and wider) had noticeable effects on the overall pressure drop. This was demonstrated in Figure 5 below, showing that the total pressure drop through the channels increased as the width-to-length ratio decreased, and as the width-to-thickness ratio increased. The final dimensions chosen were where the two lines intersected to minimize unnecessary pressure losses.



**Figure 4. Heat Exchanger Channel Dimension Ratios vs Pressure Drop**

The wall thickness was chosen based on research into industry-standard brazed plate heat exchangers.<sup>2</sup> The other important dimension was the diameter of the water distribution pipes into the plates and gathering the water after the plates. The diameter was chosen to be 2 inches, since this was the diameter of the hose feeding the heat exchanger from the buffer tank, and the diameter of the hose transporting the water to the pump. The final dimensions of the heat exchanger are shown below in Table 2.

**Table 2. Heat Exchanger Dimensions**

| Parameter                  | Value  |
|----------------------------|--------|
| Number of Channels         | 70     |
| Channel Height             | 320 mm |
| Channel Width              | 500 mm |
| Channel Thickness          | 3 mm   |
| Wall Thickness             | 0.4 mm |
| Distributing Pipe Diameter | 2 in   |
| Total Length               | 238 mm |



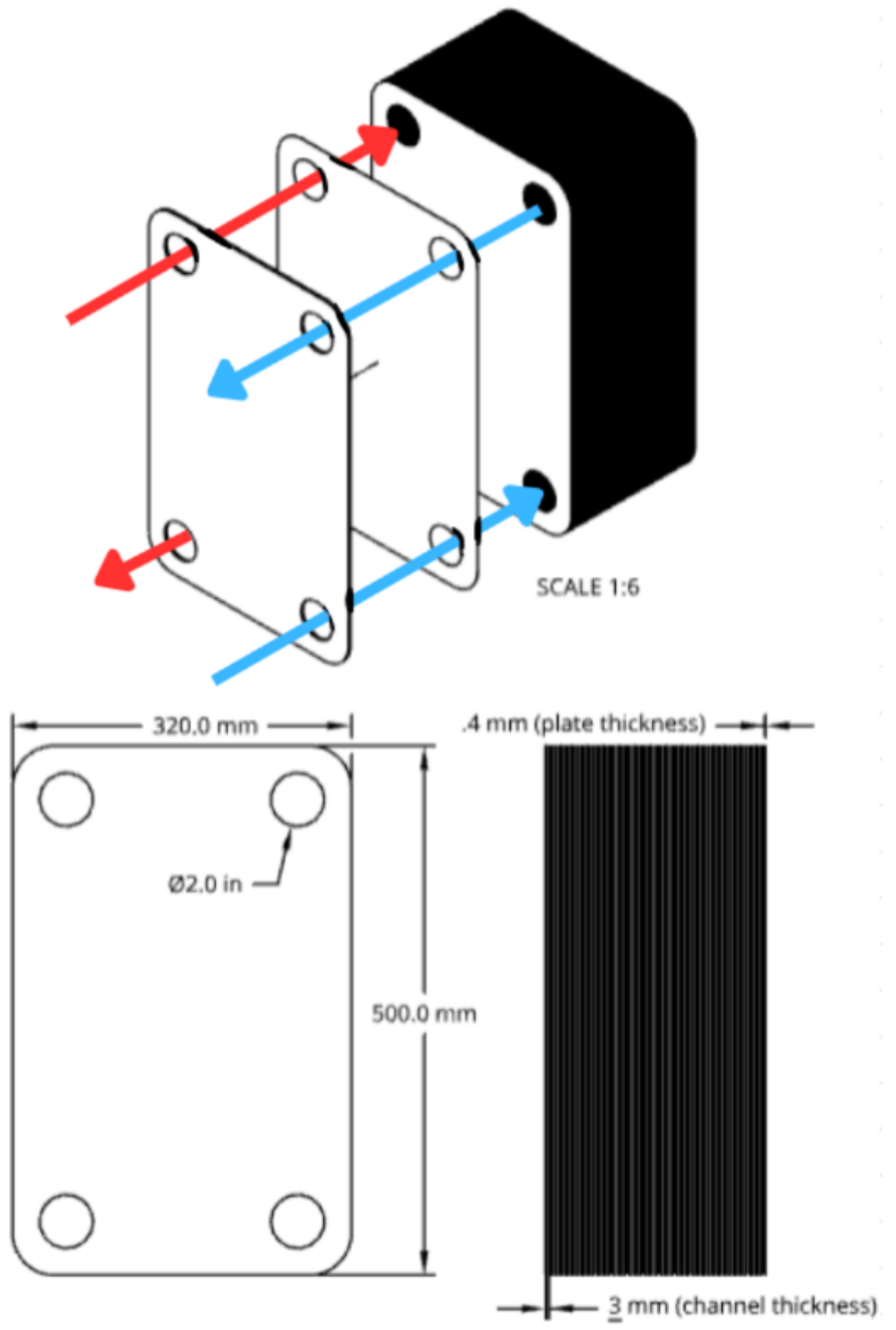
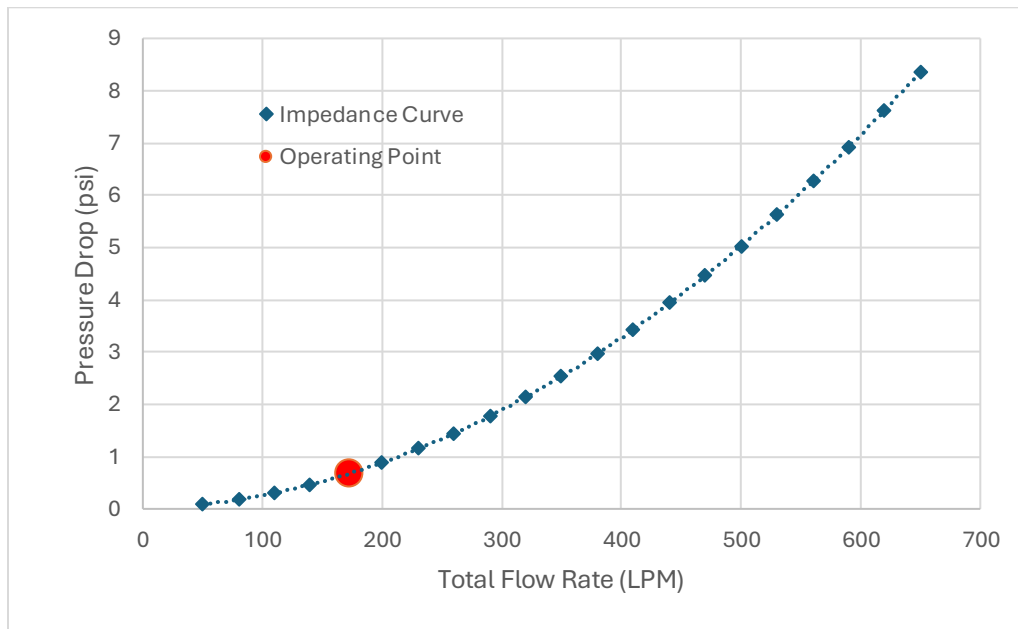
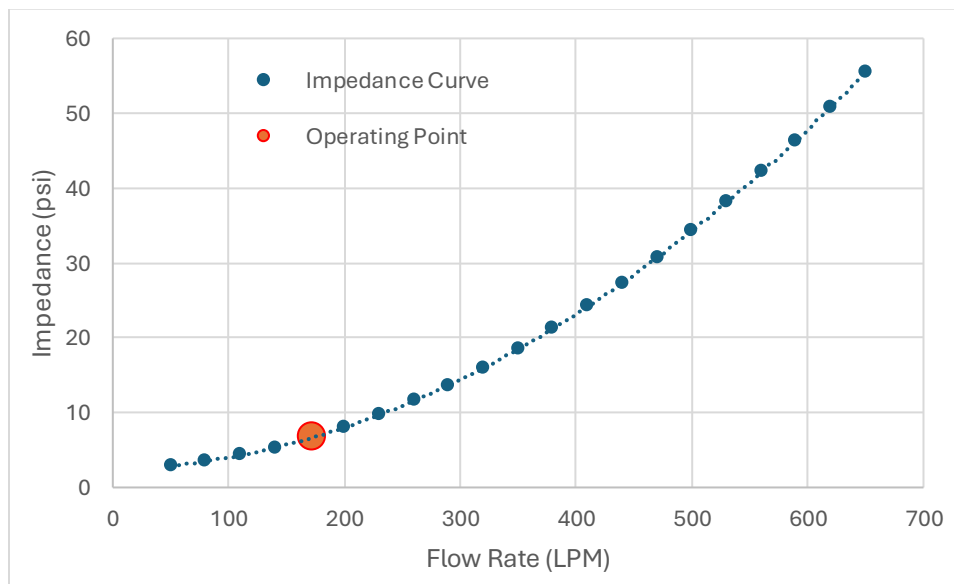


Figure 5. Heat Exchanger Model Drawing

The pressure drop equations detailed above were used to create an impedance curve for the heat exchanger from a range of 50 to 650 LPM. This was then added to the impedance curve for the rest of the system that was derived in the Interim Report. This final impedance curve was used to select a pump to move the water through the system. The impedance curves for the heat exchanger and total system are shown below in Figures 6 and 7.



**Figure 6. Overall Pressure Drop vs Flow Rate for Hot Side of HTX**



**Figure 7. Overall System Hydraulic Impedance Curve**

The final heat exchanger parameters are shown below in Table 3.

**Table 3. Heat Exchanger Design Parameters**

| Parameter                         | Value                  |
|-----------------------------------|------------------------|
| Heat Transfer Coefficient (h)     | 831 W/m <sup>2</sup> K |
| Surface Area                      | 22.4 m <sup>2</sup>    |
| NTU                               | 1.5                    |
| Hot Water Channel Velocity        | 0.127 m/s              |
| Total Heat Transfer               | 288 kW                 |
| How Water Channel Reynolds Number | 326                    |
| Effectiveness                     | .827                   |

#### **4. CDU Pump Specifications**

Once the heat exchanger geometry and associated heat load were determined, the required plant flow rate needed to achieve the CDU's thermal design was established. The total pressure drop across the entire system was then calculated. The chosen flow rate determined pressure loss through the HTX, and upon summing the pressure lost through the heat exchanger with the rest of the system, the impedance curve was generated. With the operating conditions defined, multiple pump vendor performance datasets were analyzed to identify a suitable unit. The final selection prioritized a pump whose curve closely intersected the impedance curve at or near its peak efficiency operating point. Selecting the pump in this manner ensured hydraulically stable operation, minimized energy consumption, and optimal thermal performance of the cooling system.

The impedance for choosing the pump was measured in feet of pump head. The conversion from psi to feet of head is shown below.

$$\text{Ft of Head} = 2.31 * \text{psi} \quad (13)$$

The ideal operating point for the system was calculated to be 15.43 feet of head at 172.8 LPM, or 10.32 m<sup>3</sup>/hour. Figures 8 and 9 below show the system impedance curve overlayed on a pump curve, and then the final pump curve of the selected pump.

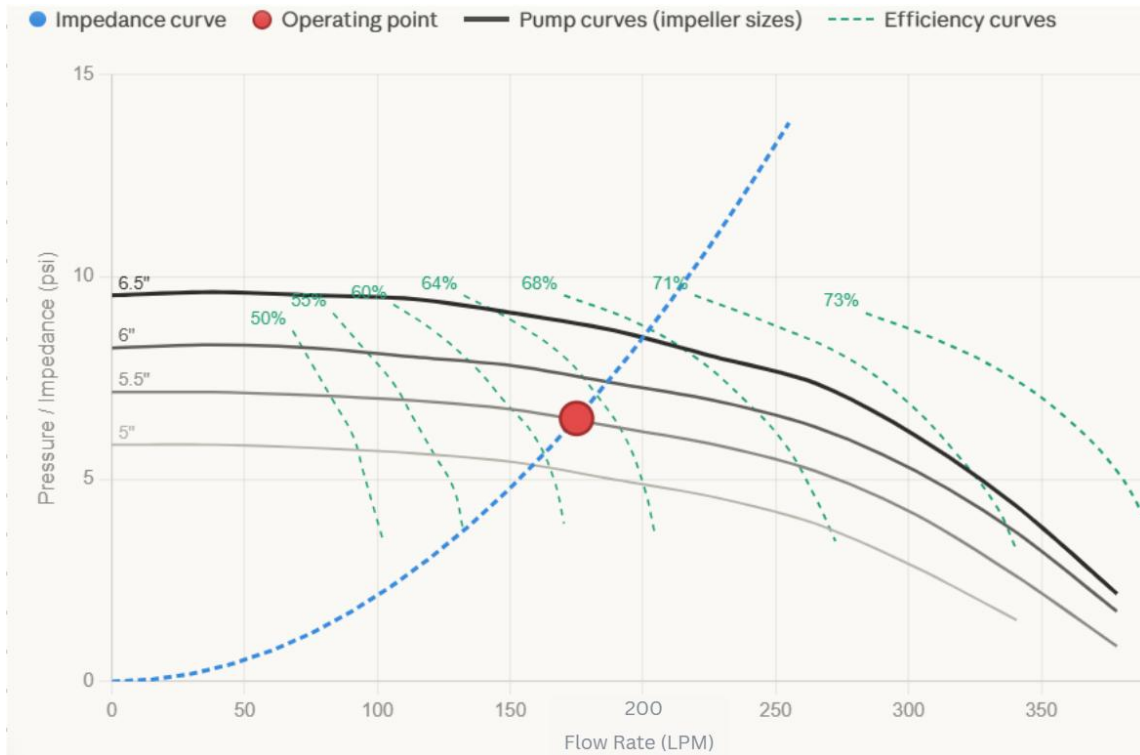


Figure 8. Pump Curve overlayed on System Impedance Curve

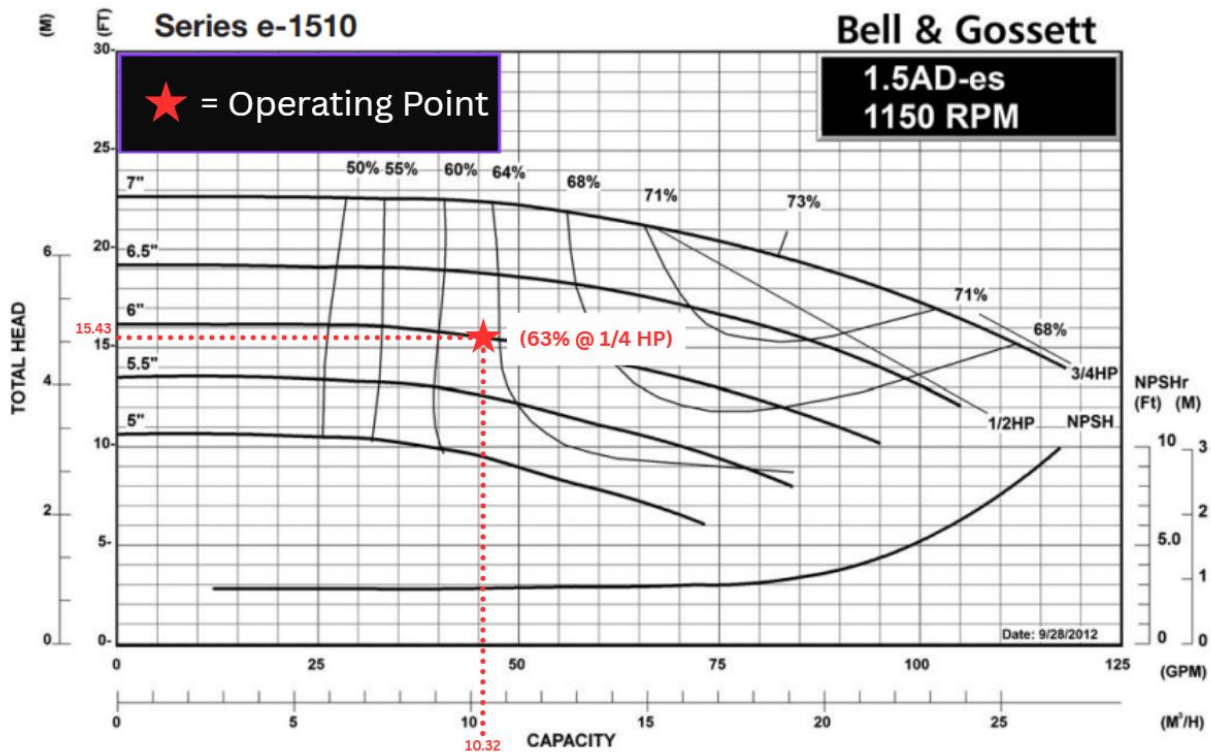


Figure 9. Operating Point on Manufacturer's Pump Curve

The selected pump was a Bell & Gossett Series e-1510 pump, which operated at 1150 RPM. The operating point for this system would have been at 15.43 feet of head and 10.32 m<sup>3</sup>/hour of water, which equated to 172.8 LPM and around 6.6 psi of total pressure drop. The pump operated at 0.25 HP.

## 5. Conclusions and Recommendations

In this final report, a brazed plate heat exchanger was designed to cool the water used in a computer server rack. A pump was then selected based on the total system impedance to produce the necessary head to overcome the losses of the system. The heat exchanger was assumed to be built with smooth plates, which produced a conservative sizing estimate for the heat exchanger. Most industry heat exchangers of this type use Chevron ribbed plates to increase the heat transfer coefficient, which in turn requires less total surface area.

If this project were to be expanded upon, the recommendation would be to design the heat exchanger without the smooth plates. Research led to a paper that directly compared smooth plates and ribbed plates in heat exchangers.<sup>3</sup> Their conclusions demonstrated that the ribbed plates increased heat transfer coefficients by around 300%. With this change, the required surface area of the heat exchanger in the system designed in this Final Report decreased by 350%. This change would have made the exact value of the heat transfer coefficient more difficult to calculate, but would have saved a lot of material, and therefore cost, in the final design.

## References

- (1) *Pipe and Tube System Components - Minor (Dynamic) Loss Coefficients*. [https://www.engineeringtoolbox.com/minor-loss-coefficients-pipes-d\\_626.html](https://www.engineeringtoolbox.com/minor-loss-coefficients-pipes-d_626.html) (accessed 2026-03-25).
- (2) *Brazed Plate Heat Exchangers | HRS Heat Exchangers*. <https://www.hrs-heatexchangers.com/heat-exchangers/plate-heat-exchangers/plate-heat-exchangers-brazed/> (accessed 2026-03-25).
- (3) *An experimental study of the flow and heat transfer between enhanced heat transfer plates for PHEs - ScienceDirect*. <https://www.sciencedirect.com/science/article/pii/S0894177710001007> (accessed 2026-03-25).

## Appendix A

### A-1. Effectiveness-NTU Chart for Counter Flow Heat Exchanger

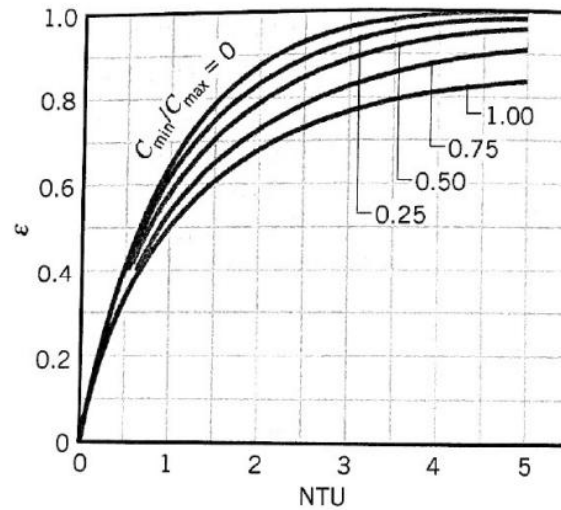


FIGURE 11.11 Effectiveness of a counterflow heat exchanger (Equation 11.29).

### A-2. Moody Chart for Pipe Friction Factors

